SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL

AI/ML 1st YEAR - MID SEM TEST-1 (MT-1) MODEL PAPERS

5 Subjects | 20-Marks Pattern | Real Questions + % Chance

Ye kya hai: SIRT ke actual MT-1 papers college-internal hote hain (publicly available nahi). Isliye RGPV ke 6 asli papers (Jun 2023 - Jun 2026) me se Mid-Sem-1 zone (Unit 1-2) ke questions uthakar 5 model papers banaye hain - same pattern, same level, % chance ke saath.

SIRT MT-1 PATTERN: Time 1:30 hr | Max Marks 20 | Koi bhi 4 out of 5 questions attempt | Har Q: (a) = 3 marks + (b) = 2 marks | Semester me 2 mid-sems hote hain, best score count hota hai | Papers CO-mapped (Unit 1-2 syllabus)

Asli SIRT papers kahan milenge: (1) SIRT ERP/student portal (2) Class CR ya subject faculty se previous MT-1 (3) Seniors - 2nd year ke paas pichle saal ke papers hote hain (4) Department notice board - MT-1 se pehle syllabus-scope notification aata hai.

PAPER 1 - BT-201 ENGINEERING PHYSICS | Unit 1: Quantum Mechanics + Unit 2: Wave Optics

Q	QUESTION ((a) = 3 marks + (b) = 2 marks)	SOURCE & % CHANCE
Q1	(a) (a) Particle in 1D box - energy eigenvalues + wavefunction ka pura derivation (b) (b) Wave function ki physical significance	RGPV: saare 6 papers - 100% Jun 2025
Q2	(a) (a) Time-Independent Schrödinger Equation derive karo (b) (b) Heisenberg uncertainty principle - statement + elementary proof	Dec 2025/Jun 2026 - 100% 4/6 papers (67%)
Q3	(a) (a) Newton's rings - dark & bright fringes ke diameter ka expression (b) (b) Numerical: λ=6000Å, 10th dark ring dia 0.50 cm - R & thickness nikalo	4/6 papers (67%) Dec 2025 actual
Q4	(a) (a) Rayleigh criterion of resolution + grating ki resolving power (b) (b) Fraunhofer single-slit - first subsidiary max = 4.5% of principal	4/6 papers (67%) Dec 2025 actual
Q5	(a) (a) Michelson interferometer - working + fringe shift derivation (b) (b) YDSE - fringe width ka formula	3/6 (50%, due) Jun 2023

PAPER 2 - BT-102 MATHEMATICS-I | Unit 1: Differential Calculus + Unit 2: Integral Calculus (Beta-Gamma)

Q	QUESTION ((a) = 3 marks + (b) = 2 marks)	SOURCE & % CHANCE
Q1	(a) (a) Rolle's theorem verify: $f(x) = e^x \sin x$, $x \in [0, 2\pi]$ (b) (b) Lagrange's theorem verify: $f(x) = x^2 + 2x$, $x \in [-2, 0]$	Jun 2026 actual (80%) Dec 2023
Q2	(a) (a) $u = \log(x^3+y^3+z^3-3xyz)$, prove $(\partial x+\partial y+\partial z)^2 u = -9/(x+y+z)^2$ (b) (b) $u = \tan^{-1}((x^3+y^3)/(x-y))$, prove $x\cdot u_x + y\cdot u_y = \sin 2u$	Dec 2025 + Jun 2026 SAME - 100% Jun 2025
Q3	(a) (a) x^3+y^3-3axy ke extreme values nikalo (b) (b) Open-top box, volume 32 cu.ft - least material wale dimensions	100% - saare papers Jun 2026 actual
Q4	(a) (a) $f(x) = \sin x$ ko $x = \pi/3$ ke about Taylor series me expand karo (b) (b) $e^x \cos y$ ka expansion (0,0) ke near - first 6 terms	Jun 2026 actual Dec 2023 (100% topic)
Q5	(a) (a) Prove: $B(m,n) = \Gamma(m)\Gamma(n)/\Gamma(m+n)$ (b) (b) Evaluate: $\int_0^1 x^4(1-\sqrt{x})^5\,dx$ ya $\int_0^1 x^2(1-x)^3\,dx$	100% Dec 2025/Dec 2024 actual

PAPER 3 - BT-203 BASIC MECHANICAL ENGINEERING | Unit 1: Materials + Instruments, Unit 2: Manufacturing

Q	QUESTION ((a) = 3 marks + (b) = 2 marks)	SOURCE & % CHANCE
Q1	(a) (a) Engineering materials classification + alloys ke applications (b) (b) Define: Toughness, Brittleness, Ductility, Tensile strength	100% (5/5 papers) Jun 2025 actual
Q2	(a) (a) Tensile test procedure + ductile material ka stress-strain curve (b) (b) Numerical: d=10mm, gauge 50mm, ultimate 60kN, breaking 40kN, rupture d=8mm - stresses + % elongation	100% Dec 2023 actual
Q3	(a) (a) Hardness + Brinell hardness test detail me (b) (b) Hooke's law derive + modulus of elasticity	Dec 2023 (due - chance hai) Jun 2026
Q4	(a) (a) Vernier caliper - construction, principle, least count (b) (b) Sine bar - construction + taper measurement	100% Jun 2025/Dec 2023 actual
Q5	(a) (a) Lathe machine ke main parts - neat sketch (b) (b) Drilling machine + koi 2 operations	100% Jun 2025/Dec 2024

PAPER 4 - BT-204 BASIC CIVIL + MECHANICS | Unit 1: Materials + Building Components, Unit 2: Surveying

Q	QUESTION ((a) = 3 marks + (b) = 2 marks)	SOURCE & % CHANCE
Q1	(a) (a) Concrete ki workability + affecting factors (b) (b) Compaction & curing ke methods	Dec 2023 actual Dec 2024 (100% zone)
Q2	(a) (a) Bricks - manufacturing + good brick properties & lab tests (b) (b) Stones ki characteristics + testing methods	Dec 2025 (80%) Jun 2025
Q3	(a) (a) Staircase ke types + suitability (b) (b) Spread footing & RCC footing - sketches	100% zone Dec 2025/Jun 2026
Q4	(a) (a) Numerical (Rise-Fall): 12 readings @20m, instrument shift after 4th & 8th, last on BM RL 125.250 - sab RL + check (b) (b) HI method: BS=1.425, FS=2.165, BM RL=100.000 - next RL	Dec 2024 + Jun 2026 actual (80% - numerical pakka)
Q5	(a) (a) Contour - properties + contouring methods (b) (b) Remote sensing applications + GPS positioning principle	4/5 (80%) 100% zone (Jun 2026)

PAPER 5 - BT-205 BASIC COMPUTER ENGINEERING | Unit 1: Fundamentals + Unit 2: OS, Algorithms, OOP

Q	QUESTION ((a) = 3 marks + (b) = 2 marks)	SOURCE & % CHANCE
Q1	(a) (a) Computer classification (size/functionality) - super vs mainframe vs embedded (b) (b) 4 fields me computers ka role: E-business, Healthcare, GIS, Meteorology	Jun 2025 + Jun 2026 actual (80%)
Q2	(a) (a) System vs application software - difference + examples (b) (b) Primary vs secondary memory + memory hierarchy	3/5 (60%) 80%
Q3	(a) (a) OS ke main functions + file/process/memory management (b) (b) Kernel ka concept	4/5 (80%) Jun 2025
Q4	(a) (a) Good algorithm ki characteristics + flowchart: largest of 3 numbers (b) (b) Flowcharts ki limitations	3 BAAR REPEAT - 100%! Dec24+Jun25+Jun26
Q5	(a) (a) OOP characteristics + POP limitations kaise overcome karta hai (b) (b) Paradigms compare: procedural vs OOP vs functional	100% (5/5) Dec 2023

MT-1 NIGHT-BEFORE STRATEGY - sirf ye karo to bhi 14-16/20 pakke!

1. PHYSICS: Particle-in-box derivation + Newton's rings numerical - in 2 me se 2 questions pakka milenge.
2. MATHS: u=log(...) wala derivative + maxima x^3+y^3-3axy + Taylor series - teeno ratt lo (100% category).
3. BME: Stress-strain curve + vernier/sine bar + lathe sketch - Q1/Q4/Q5 yahin se aayenge.
4. BCEM: RL numerical (Rise-Fall + HI dono) + bricks + contours.
5. BCE: Largest-of-3 flowchart likhna AANA hi chahiye (3 baar repeat!) + OOP characteristics + OS functions.

FORMULA: 4 questions attempt karne hain. 100%/80% wale topics se har paper me 4+ strong options mil jayenge. MT-1 se pehle faculty jo scope batayegi (kaunse units) wahi confirm kar lena - aur seniors se pichle saal ka SIRT MT-1 zaroor mangwao, pattern repeat hota hai. All the best!

Base: RGPV actual papers Jun 2023 - Jun 2026 (MT-1 zone = Unit 1-2). SIRT MT-1 me faculty CO-mapped questions banati hai, lekin source wahi RGPV syllabus hai - isliye ye high-probability questions hain. Ye model practice papers hain, official SIRT question papers nahi. Compiled: October 2026.